PADHAKU
Mathematics · Class 10 Mathematics



CHAPTER 8

Introduction to Trigonometry

NCERT Exercise Solutions

In-text questions	11 Q
Exercise 8.1	11 Q
Exercise 8.2	11 Q
Exercise 8.3	16 Q

✨ Try each question yourself first, then check the answer. 💡 answers are samples: write your own.

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In-text questions

8.2 (Fig. 8.5) **Define the trigonometric ratios for angle C in the right triangle ABC (right-angled at B).**

✔ ANSWER

For ∠C, the opposite side is AB, the adjacent side is BC, the hypotenuse is AC.

sin C = AB/AC, cos C = BC/AC, tan C = AB/BC

cosec C = AC/AB, sec C = AC/BC, cot C = BC/AB

8.2 (Fig. 8.6) **How will the trigonometric ratios of ∠A in Δ PAM differ from those of ∠A in Δ CAB or in Δ QAN?**

✔ ANSWER

They do not differ at all.

Δ PAM ~ Δ CAB ~ Δ QAN (AA similarity: common ∠A and a right angle in each), so corresponding sides are proportional: MP/AP = BC/AC = NQ/AQ, etc.

The trigonometric ratios of an angle depend only on the angle, not on the size of the triangle.

8.2 (Why?) **AB = ±2√2 k. Why is AB not −2√2 k?**

✔ ANSWER

AB is the length of a side of a triangle, and a length is always positive.

So AB = 2√2 k (the negative value is rejected).

Example 5 (Why?) **Why is $(1 + PQ)^2 = OP^2 + PQ^2$? Why is $1 + 2PQ = 7^2$?**

✔ ANSWER

Given $OQ - PQ = 1$, so $OQ = 1 + PQ$; Pythagoras gives $OQ^2 = OP^2 + PQ^2$, i.e. $(1 + PQ)^2 = OP^2 + PQ^2$.

Expanding: $1 + 2PQ + PQ^2 = OP^2 + PQ^2$. PQ^2 cancels and OP = 7 cm, so **$1 + 2PQ = 7^2 = 49$**.

8.3 (Why?) **In Δ ABC right-angled at B with ∠A = ∠C = 45°, why is BC = AB?**

✔ ANSWER

Sides opposite to equal angles of a triangle are equal.

BC is opposite ∠A and AB is opposite ∠C, and ∠A = ∠C, so **BC = AB**.

8.3 (Why?) **In equilateral Δ ABC with AD ⊥ BC, why is Δ ABD ≅ Δ ACD?**

✔ ANSWER

AB = AC (sides of an equilateral triangle), AD = AD (common), ∠ADB = ∠ADC = 90°.

Δ ABD ≅ Δ ACD by the RHS congruence rule.
Hence BD = DC and ∠BAD = ∠CAD (CPCT).

8.3 (Why?) **Why are cot 0° and cosec 0° not defined?**

✔ ANSWER

cot 0° = cos 0°/sin 0° = 1/0 and cosec 0° = 1/sin 0° = 1/0.

Division by zero is not defined, so cot 0° and cosec 0° are not defined.

8.3 **Find the other trigonometric ratios of 90° (given sin 90° = 1, cos 90° = 0).**

✔ ANSWER

tan 90° = sin 90°/cos 90° = 1/0 → **not defined**

cot 90° = cos 90°/sin 90° = 0/1 = **0**

sec 90° = 1/cos 90° = 1/0 → **not defined**

cosec 90° = 1/sin 90° = 1/1 = **1**

Example 6 (Why?) **Why is sin 30° = AB/AC?**

✔ ANSWER

In Δ ABC right-angled at B, ∠C = 30°. The side opposite ∠C is AB and the hypotenuse is AC.

sin C = sin 30° = opposite/hypotenuse = AB/AC.

Example 7 (Why?) **If ∠PRQ = 30°, why is ∠QPR = 60°?**

✔ ANSWER

In Δ PQR, ∠Q = 90°. Angle sum: ∠P + ∠Q + ∠R = 180°.

∠QPR = 180° − 90° − 30° = 60°.

Example 8 (Why?) **Why does sin (A − B) = ½ give A − B = 30°, and cos (A + B) = ½ give A + B = 60°?**

✔ ANSWER

From Table 8.1, sin 30° = ½ and cos 60° = ½.

A − B and A + B are angles between 0° and 90°, and in this range each value of sine or cosine comes from only one angle.

So A − B = 30° and A + B = 60°.

Exercise 8.1

1 **In Δ ABC, right-angled at B, AB = 24 cm, BC = 7 cm. Determine: (i) sin A, cos A (ii) sin C, cos C**

✔ ANSWER

By Pythagoras theorem: $AC^2 = AB^2 + BC^2 = 24^2 + 7^2 = 576 + 49 = 625$, so AC = 25 cm.

(i) For ∠A: opposite = BC, adjacent = AB.

sin A = BC/AC = 7/25, cos A = AB/AC = 24/25

(ii) For ∠C: opposite = AB, adjacent = BC.

sin C = AB/AC = 24/25, cos C = BC/AC = 7/25

2 **In Fig. 8.13 (Δ PQR right-angled at Q, PQ = 12 cm, PR = 13 cm), find tan P − cot R.**

✔ ANSWER

$QR^2 = PR^2 - PQ^2 = 13^2 - 12^2 = 169 - 144 = 25$, so QR = 5 cm.

tan P = QR/PQ = 5/12

cot R = QR/PQ = 5/12 (for ∠R, adjacent = QR, opposite = PQ)

tan P − cot R = 5/12 − 5/12 = 0

3 **If sin A = 3/4, calculate cos A and tan A.**

✔ ANSWER

sin A = BC/AC = 3/4. Let BC = 3k, AC = 4k (right angle at B).

$AB^2 = AC^2 - BC^2 = 16k^2 - 9k^2 = 7k^2$, so AB = √7 k.

cos A = AB/AC = √7/4

tan A = BC/AB = 3/√7

4 **Given 15 cot A = 8, find sin A and sec A.**

✔ ANSWER

cot A = 8/15 = AB/BC. Let AB = 8k, BC = 15k (right angle at B).

$AC^2 = 8^2k^2 + 15^2k^2 = 64k^2 + 225k^2 = 289k^2$, so AC = 17k.

sin A = BC/AC = 15/17

sec A = AC/AB = 17/8

5 **Given sec θ = 13/12, calculate all other trigonometric ratios.**

✔ ANSWER

sec θ = hypotenuse/adjacent = 13/12. Let hypotenuse = 13k, adjacent = 12k.

opposite^2 = $169k^2 - 144k^2 = 25k^2$, so opposite = 5k.

sin θ = 5/13, cos θ = 12/13, tan θ = 5/12, cot θ = 12/5, cosec θ = 13/5

6 **If ∠A and ∠B are acute angles such that cos A = cos B, then show that ∠A = ∠B.**

✔ ANSWER

1. Take Δ ABC with CD ⊥ AB (D on AB). Then ∠A and ∠B are the acute angles of the right triangles ADC and BDC.

2. cos A = AD/AC and cos B = BD/BC.

3. cos A = cos B ⇒ AD/AC = BD/BC ⇒ AD/BD = AC/BC = m (say). So AC = m·BC and AD = m·BD.

4. By Pythagoras in both triangles, $CD^2 = AC^2 - AD^2 = BC^2 - BD^2$.

5. So $m^2 \cdot BC^2 - m^2 \cdot BD^2 = BC^2 - BD^2$, i.e. $(m^2 - 1)(BC^2 - BD^2) = 0$.

6. $BC^2 - BD^2 = CD^2 \neq 0$, so $m^2 = 1$, i.e. $m = 1$ (lengths are positive). Hence AC = BC.

7. In Δ ABC, AC = BC, so the angles opposite them are equal: **$\angle A = \angle B$**.

7 **If $\cot\theta = 7/8$, evaluate: (i) $(1 + \sin\theta)(1 - \sin\theta) / ((1 + \cos\theta)(1 - \cos\theta))$ (ii) $\cot^2\theta$**

✔ ANSWER

$\cot\theta$ = adjacent/opposite = 7/8. Let adjacent = 7k, opposite = 8k; hypotenuse = $\sqrt{(49 + 64)}\,k = \sqrt{113}\,k$.

$\sin\theta = 8/\sqrt{113}$, $\cos\theta = 7/\sqrt{113}$.

(i) $(1 + \sin\theta)(1 - \sin\theta) / ((1 + \cos\theta)(1 - \cos\theta))$ $= (1 - \sin^2\theta)/(1 - \cos^2\theta) = \cos^2\theta / \sin^2\theta$

$= (49/113) \div (64/113) =$ **49/64**

(ii) $\cot^2\theta = (7/8)^2 =$ **49/64**

8 **If $3\cot A = 4$, check whether $(1 - \tan^2 A)/(1 + \tan^2 A) = \cos^2 A - \sin^2 A$ or not.**

✔ ANSWER

$\cot A = 4/3$, so $\tan A = 3/4$.

LHS = $(1 - 9/16)/(1 + 9/16) = (7/16)/(25/16) = 7/25$

For RHS: AB = 4k, BC = 3k, AC = $\sqrt{(16 + 9)}\,k = 5k$. $\cos A = 4/5$, $\sin A = 3/5$.

RHS = $16/25 - 9/25 = 7/25$

LHS = RHS = 7/25. Yes, the two expressions are equal.

9 **In triangle ABC, right-angled at B, if tan A = 1/√3, find the value of: (i) sin A cos C + cos A sin C (ii) cos A cos C − sin A sin C**

✔ ANSWER

tan A = BC/AB = 1/√3. Let BC = k, AB = √3 k; AC = √(k² + 3k²) = 2k.

sin A = BC/AC = 1/2, cos A = AB/AC = √3/2

sin C = AB/AC = √3/2, cos C = BC/AC = 1/2

(i) sin A cos C + cos A sin C = (1/2)(1/2) + (√3/2)(√3/2) = 1/4 + 3/4 = **1**

(ii) cos A cos C − sin A sin C = (√3/2)(1/2) − (1/2)(√3/2) = √3/4 − √3/4 = **0**

10 **In Δ PQR, right-angled at Q, PR + QR = 25 cm and PQ = 5 cm. Determine the values of sin P, cos P and tan P.**

✔ ANSWER

Let QR = x cm. Then PR = (25 − x) cm.

Pythagoras: $PR^2 = PQ^2 + QR^2 \Rightarrow (25 - x)^2 = 25 + x^2$

$625 - 50x + x^2 = 25 + x^2 \Rightarrow 50x = 600 \Rightarrow x = 12$

So QR = 12 cm, PR = 13 cm.

sin P = QR/PR = 12/13, cos P = PQ/PR = 5/13, tan P = QR/PQ = 12/5

11 **State whether the following are true or false. Justify your answer.**

(i) The value of tan A is always less than 1.

(ii) sec A = 12/5 for some value of angle A.

(iii) cos A is the abbreviation used for the cosecant of angle A.

(iv) cot A is the product of cot and A.

(v) sin θ = 4/3 for some angle θ.

✔ ANSWER

(i) **False.** tan A = opposite/adjacent can be more than 1, e.g. tan 60° = √3 ≈ 1.73.

(ii) **True.** sec A = hypotenuse/adjacent is always ≥ 1, and 12/5 = 2.4 > 1, so a right triangle with hypotenuse 12 and adjacent side 5 gives sec A = 12/5.

(iii) **False.** cos A is the abbreviation of cosine of angle A; cosecant is written cosec A.

(iv) **False.** cot A is one symbol meaning the cotangent of angle A; 'cot' alone has no meaning, so it is not a product.

(v) **False.** sin θ = opposite/hypotenuse and the hypotenuse is the longest side, so sin θ ≤ 1; but 4/3 > 1.

Exercise 8.2

1 (i) **Evaluate: sin 60° cos 30° + sin 30° cos 60°**

✔ ANSWER

= (√3/2)(√3/2) + (1/2)(1/2)

= 3/4 + 1/4 = **1**

1 (ii) Evaluate: 2 tan² 45° + cos² 30° − sin² 60°

✔ ANSWER

= 2(1)² + (√3/2)² − (√3/2)²

= 2 + 3/4 − 3/4 = **2**

1 (iii) Evaluate: cos 45° / (sec 30° + cosec 30°)

✔ ANSWER

= (1/√2) ÷ (2/√3 + 2)

= (1/√2) ÷ ((2 + 2√3)/√3) = √3 / (√2 (2 + 2√3))

= √3 / (2√2 (1 + √3))

Rationalise with (√3 − 1): = √3(√3 − 1) / (2√2 (3 − 1)) = (3 − √3)/(4√2)

Multiply by √2/√2: = (3√2 − √6)/8

= (3√2 − √6)/8

1 (iv) Evaluate: (sin 30° + tan 45° − cosec 60°) / (sec 30° + cos 60° + cot 45°)

✔ ANSWER

Numerator = 1/2 + 1 − 2/√3 = 3/2 − 2/√3

Denominator = 2/√3 + 1/2 + 1 = 3/2 + 2/√3

Multiply both by 2√3: = (3√3 − 4)/(3√3 + 4)

Rationalise with (3√3 − 4): = (3√3 − 4)² / ((3√3)² − 4²) = (27 − 24√3 + 16)/(27 − 16)

= (43 − 24√3)/11

1 (v) **Evaluate: (5 cos² 60° + 4 sec² 30° − tan² 45°) / (sin² 30° + cos² 30°)**

✔ ANSWER

Numerator = 5(1/2)² + 4(2/√3)² − 1² = 5/4 + 16/3 − 1 = (15 + 64 − 12)/12 = 67/12

Denominator = (1/2)² + (√3/2)² = 1/4 + 3/4 = 1

= 67/12

2 (i) **2 tan 30° / (1 + tan² 30°) = (A) sin 60° (B) cos 60° (C) tan 60° (D) sin 30°**

✔ ANSWER

(A) sin 60°

2 tan 30° / (1 + tan² 30°) = (2/√3) ÷ (1 + 1/3) = (2/√3) × (3/4) = √3/2 = sin 60°.

2 (ii) **(1 − tan² 45°) / (1 + tan² 45°) = (A) tan 90° (B) 1 (C) sin 45° (D) 0**

✔ ANSWER

(D) 0

tan 45° = 1, so (1 − 1)/(1 + 1) = 0/2 = 0.

2 (iii) **sin 2A = 2 sin A is true when A = (A) 0° (B) 30° (C) 45° (D) 60°**

✔ ANSWER

(A) 0°

For A = 0°: sin 0° = 0 and 2 sin 0° = 0, equal. For 30°: sin 60° = √3/2 but 2 sin 30° = 1; similarly 45° and 60° do not work.

2 (iv) **2 tan 30° / (1 − tan² 30°) = (A) cos 60° (B) sin 60° (C) tan 60° (D) sin 30°**

✔ ANSWER

(C) tan 60°

(2/√3) ÷ (1 − 1/3) = (2/√3) × (3/2) = 3/√3 = √3 = tan 60°.

3 **If tan (A + B) = √3 and tan (A − B) = 1/√3; 0° < A + B ≤ 90°; A > B, find A and B.**

✔ ANSWER

tan (A + B) = √3 = tan 60°, so A + B = 60° ... (1)

tan (A − B) = 1/√3 = tan 30°, so A − B = 30° ... (2)

Adding (1) and (2): 2A = 90°, so A = 45°.

From (1): B = 60° − 45° = 15°.

A = 45°, B = 15°

4 **State whether the following are true or false. Justify your answer.**

(i) sin (A + B) = sin A + sin B.

(ii) The value of sin θ increases as θ increases.

(iii) The value of cos θ increases as θ increases.

(iv) sin θ = cos θ for all values of θ.

(v) cot A is not defined for A = 0°.

✔ ANSWER

(i) **False.** Take A = B = 30°: sin 60° = √3/2, but sin 30° + sin 30° = 1/2 + 1/2 = 1. Not equal.

(ii) **True.** sin 0° = 0, sin 30° = 1/2, sin 45° = 1/√2, sin 60° = √3/2, sin 90° = 1: the value keeps increasing from 0 to 1.

(iii) **False.** cos 0° = 1, cos 30° = √3/2, cos 45° = 1/√2, cos 60° = 1/2, cos 90° = 0: the value decreases as θ increases.

(iv) **False.** It is true only for θ = 45°. For example, sin 30° = 1/2 but cos 30° = √3/2.

(v) **True.** cot 0° = cos 0°/sin 0° = 1/0, and division by zero is not defined.

Exercise 8.3

1 **Express the trigonometric ratios sin A, sec A and tan A in terms of cot A.**

✔ ANSWER

$\text{cosec}^2 A = 1 + \cot^2 A \Rightarrow \text{cosec } A = \sqrt{(1 + \cot^2 A)}$

$\sin A = 1/\text{cosec } A = 1/\sqrt{(1 + \cot^2 A)}$

$\tan A = 1/\cot A$

$\sec^2 A = 1 + \tan^2 A = 1 + 1/\cot^2 A = (\cot^2 A + 1)/\cot^2 A$

sec A = √(1 + cot² A) / cot A

2 **Write all the other trigonometric ratios of ∠A in terms of sec A.**

✔ ANSWER

cos A = 1/sec A

sin² A = 1 − cos² A = 1 − 1/sec² A = (sec² A − 1)/sec² A ⇒ **sin A = √(sec² A − 1) / sec A**

tan² A = sec² A − 1 ⇒ **tan A = √(sec² A − 1)**

cot A = 1/tan A = 1/√(sec² A − 1)

cosec A = 1/sin A = sec A / √(sec² A − 1)

3 (i) **9 sec² A − 9 tan² A = (A) 1 (B) 9 (C) 8 (D) 0**

✔ ANSWER

(B) 9

9 sec² A − 9 tan² A = 9 (sec² A − tan² A) = 9 × 1 = 9.

3 (ii) **(1 + tan θ + sec θ)(1 + cot θ − cosec θ) = (A) 0 (B) 1 (C) 2 (D) −1**

✔ ANSWER

(C) 2

Write in sin and cos: (cos θ + sin θ + 1)/cos θ × (sin θ + cos θ − 1)/sin θ

= [(sin θ + cos θ)² − 1] / (sin θ cos θ) = (sin² θ + cos² θ + 2 sin θ cos θ − 1)/(sin θ cos θ)

= 2 sin θ cos θ / (sin θ cos θ) = 2.

3 (iii) **(sec A + tan A)(1 − sin A) = (A) sec A (B) sin A (C) cosec A (D) cos A**

✔ ANSWER

(D) cos A

(1/cos A + sin A/cos A)(1 − sin A) = (1 + sin A)(1 − sin A)/cos A = (1 − sin² A)/cos A = cos² A/cos A = cos A.

3 (iv) **(1 + tan² A)/(1 + cot² A) = (A) sec² A (B) −1 (C) cot² A (D) tan² A**

✔ ANSWER

(D) tan² A

(1 + tan² A)/(1 + cot² A) = sec² A / cosec² A = (1/cos² A) × sin² A = sin² A/cos² A = tan² A.

4 (i) **Prove: (cosec θ − cot θ)² = (1 − cos θ)/(1 + cos θ)**

✔ ANSWER

LHS = (1/sin θ − cos θ/sin θ)² = (1 − cos θ)²/sin² θ

= (1 − cos θ)²/(1 − cos² θ) [sin² θ = 1 − cos² θ]

= (1 − cos θ)² / [(1 − cos θ)(1 + cos θ)]

= (1 − cos θ)/(1 + cos θ) = RHS. **Proved.**

4 (ii) **Prove: $\cos A/(1+\sin A) + (1+\sin A)/\cos A = 2\sec A$**

✔ ANSWER

LHS = $[\cos^2 A + (1+\sin A)^2] / [(1+\sin A)\cos A]$

= $[\cos^2 A + 1 + 2\sin A + \sin^2 A] / [(1+\sin A)\cos A]$

= $(2 + 2\sin A) / [(1+\sin A)\cos A]$ $[\cos^2 A + \sin^2 A = 1]$

= $2(1+\sin A) / [(1+\sin A)\cos A] = 2/\cos A = 2\sec A$ = RHS. **Proved.**

4 (iii) **Prove: $\tan\theta/(1-\cot\theta) + \cot\theta/(1-\tan\theta) = 1 + \sec\theta\,\text{cosec}\,\theta$**

✔ ANSWER

$\tan\theta/(1-\cot\theta) = (\sin\theta/\cos\theta) \div ((\sin\theta - \cos\theta)/\sin\theta) = \sin^2\theta / [\cos\theta(\sin\theta - \cos\theta)]$

$\cot\theta/(1-\tan\theta) = (\cos\theta/\sin\theta) \div ((\cos\theta - \sin\theta)/\cos\theta) = \cos^2\theta / [\sin\theta(\cos\theta - \sin\theta)] = -\cos^2\theta / [\sin\theta(\sin\theta - \cos\theta)]$

LHS = $[\sin^3\theta - \cos^3\theta] / [\sin\theta\cos\theta(\sin\theta - \cos\theta)]$

= $(\sin\theta - \cos\theta)(\sin^2\theta + \sin\theta\cos\theta + \cos^2\theta) / [\sin\theta\cos\theta(\sin\theta - \cos\theta)]$ $[a^3 - b^3 = (a-b)(a^2 + ab + b^2)]$

= $(1 + \sin\theta\cos\theta)/(\sin\theta\cos\theta) = 1/(\sin\theta\cos\theta) + 1$

= $1 + \sec\theta\,\text{cosec}\,\theta$ = RHS. **Proved.**

4 (iv) **Prove: $(1 + \sec A)/\sec A = \sin^2 A/(1 - \cos A)$**

✔ ANSWER

LHS = $1/\sec A + 1 = \cos A + 1$

RHS = $(1 - \cos^2 A)/(1 - \cos A) = (1 - \cos A)(1 + \cos A)/(1 - \cos A) = 1 + \cos A$

LHS = RHS. **Proved.**

4 (v) **Prove: $(\cos A - \sin A + 1)/(\cos A + \sin A - 1) = \operatorname{cosec} A + \cot A$, using the identity $\operatorname{cosec}^2 A = 1 + \cot^2 A$.**

✔ ANSWER

Divide numerator and denominator by sin A:

LHS = $(\cot A - 1 + \operatorname{cosec} A)/(\cot A + 1 - \operatorname{cosec} A)$

In the numerator write $1 = \operatorname{cosec}^2 A - \cot^2 A = (\operatorname{cosec} A - \cot A)(\operatorname{cosec} A + \cot A)$:

Numerator = $(\cot A + \operatorname{cosec} A) - (\operatorname{cosec} A - \cot A)(\operatorname{cosec} A + \cot A)$

$= (\operatorname{cosec} A + \cot A)[1 - (\operatorname{cosec} A - \cot A)] = (\operatorname{cosec} A + \cot A)(\cot A + 1 - \operatorname{cosec} A)$

LHS = $(\operatorname{cosec} A + \cot A)(\cot A + 1 - \operatorname{cosec} A)/(\cot A + 1 - \operatorname{cosec} A) = \operatorname{cosec} A + \cot A$ = RHS. **Proved.**

4 (vi) **Prove: $\sqrt{((1 + \sin A)/(1 - \sin A))} = \sec A + \tan A$**

✔ ANSWER

Multiply numerator and denominator inside the root by $(1 + \sin A)$:

LHS $= \sqrt{[(1 + \sin A)^2 / ((1 - \sin A)(1 + \sin A))]} = \sqrt{[(1 + \sin A)^2 / (1 - \sin^2 A)]}$

$= \sqrt{[(1 + \sin A)^2 / \cos^2 A]} = (1 + \sin A)/\cos A$

$= 1/\cos A + \sin A/\cos A = \sec A + \tan A =$ RHS. **Proved.**

4 (vii) **Prove: $(\sin \theta - 2 \sin^3 \theta)/(2 \cos^3 \theta - \cos \theta) = \tan \theta$**

✔ ANSWER

LHS $= \sin \theta (1 - 2 \sin^2 \theta) / [\cos \theta (2 \cos^2 \theta - 1)]$

$1 - 2 \sin^2 \theta = 1 - 2(1 - \cos^2 \theta) = 2 \cos^2 \theta - 1$

LHS $= \sin \theta (2 \cos^2 \theta - 1) / [\cos \theta (2 \cos^2 \theta - 1)] = \sin \theta/\cos \theta = \tan \theta =$ RHS. **Proved.**

4 (viii) **Prove: $(\sin A + \text{cosec } A)^2 + (\cos A + \sec A)^2 = 7 + \tan^2 A + \cot^2 A$**

✔ ANSWER

LHS $= \sin^2 A + 2 \sin A \text{ cosec } A + \text{cosec}^2 A + \cos^2 A + 2 \cos A \sec A + \sec^2 A$

$= (\sin^2 A + \cos^2 A) + 2 + 2 + \text{cosec}^2 A + \sec^2 A$ [$\sin A \text{ cosec } A = 1$, $\cos A \sec A = 1$]

$= 1 + 4 + (1 + \cot^2 A) + (1 + \tan^2 A)$

$= 7 + \tan^2 A + \cot^2 A =$ RHS. **Proved.**

4 (ix) **Prove: (cosec A − sin A)(sec A − cos A) = 1/(tan A + cot A)**

✔ ANSWER

LHS = (1/sin A − sin A)(1/cos A − cos A) = [(1 − $\sin^2$ A)/sin A] × [(1 − $\cos^2$ A)/cos A]

= ($\cos^2$ A/sin A)($\sin^2$ A/cos A) = sin A cos A

RHS = 1/(sin A/cos A + cos A/sin A) = 1/[($\sin^2$ A + $\cos^2$ A)/(sin A cos A)] = sin A cos A

LHS = RHS. **Proved.**

4 (x) **Prove: (1 + $\tan^2$ A)/(1 + $\cot^2$ A) = ((1 − tan A)/(1 − cot A))^2 = $\tan^2$ A**

✔ ANSWER

First part: (1 + $\tan^2$ A)/(1 + $\cot^2$ A) = $\sec^2$ A/cosec^2 A = $\sin^2$ A/$\cos^2$ A = $\tan^2$ A

Second part: (1 − tan A)/(1 − cot A) = (1 − tan A)/(1 − 1/tan A) = (1 − tan A) tan A/(tan A − 1) = −tan A

So ((1 − tan A)/(1 − cot A))^2 = (−tan A)^2 = $\tan^2$ A

Hence both expressions equal $\tan^2$ A. **Proved.**

🎉 **Chapter 8 done! Revise the answers once more tomorrow.**